

Topology Bootcamp: Week 1

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1 Handles

In many areas of math, it is a ubiquitous technique to break down complicated objects into simple, elementary pieces. When we study the topology of 3- and 4-manifolds, these pieces are *handles*.

Definition 1.1. Let X be a smooth n -manifold with boundary, and $0 \leq k \leq n$. An n -dimensional k -handle is a copy of $D^k \times D^{n-k}$ attached to ∂X along $(\partial D^k) \times D^{n-k}$ via an embedding $f : (\partial D^k) \times D^{n-k} \rightarrow \partial X$. Indeed, $X \cup_f (D^k \times D^{n-k})$ is again a smooth manifold.¹

Example 1.1 In this bootcamp, we're mostly concerned with 3- and 4-dimensional k -handles. Here's an illustration of attaching a 3D k -handle to the boundary of B^3 for all possible k :

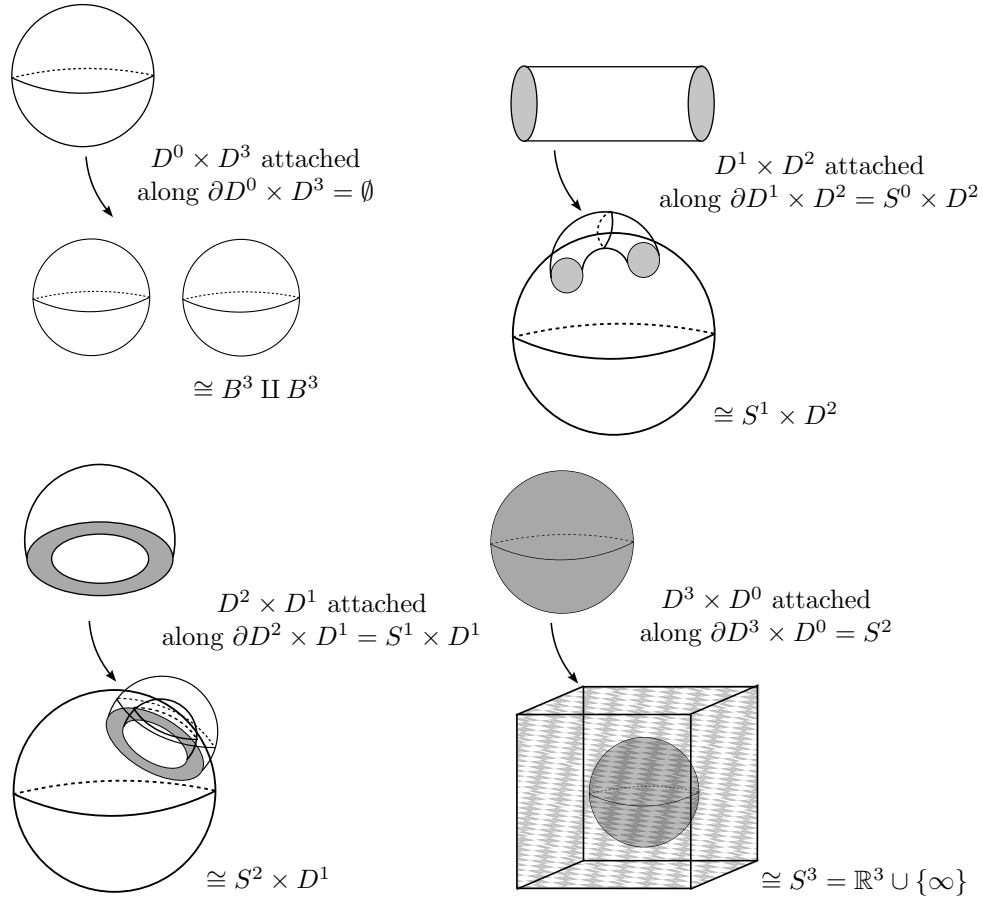


Figure 1: 3D Handle Attachments

¹The corners of the attaching region can be smoothed in a unique way, see for instance Chapter 1 of Conner-Floyd.

We will revisit how to manipulate handles in detail when we get to the section on handle calculus. For now, all we really need is the notion of *handlebodies*:

Definition 1.2. A 3-dimensional g -handlebody, denoted H_g , is the 3-manifold obtained by attaching g 1-handles to B^3 . The number g is called the *genus* of H_g . Note that $\partial(H_g)$ is the closed orientable surface of genus g .

2 Heegaard Splittings

Let M be a closed orientable 3-manifold. Intuitively, a *Heegaard splitting* is a way of decomposing M into two handlebodies of the same genus. More formally,

Definition 2.1. A *Heegaard splitting* is a decomposition $M = H \cup H'$ where H and H' are handlebodies such that $H \cap H' = \partial H = \partial H'$. Naturally, this implies that H and H' have the same genus.

Example 2.1 Gluing two copies of the solid torus $S^1 \times D^2$ via the identity map yields a manifold diffeomorphic to $S^1 \times S^2$. One can see this by starting with $S^1 \times S^2$ and decomposing it into two solid tori:

$$\begin{aligned} S^1 \times S^2 &= S^1 \times (U \cup L) \\ &= S^1 \times U \cup S^1 \times L \\ &\cong S^1 \times D^2 \cup_{\text{id}} S^1 \times D^2, \end{aligned}$$

where $U \cong L \cong D^2$ are the upper and lower hemispheres of S^2 glued along the identity on their boundary.

Example 2.2 If we instead glue along the homeomorphism $f : S^1 \times S^1 \rightarrow S^1 \times S^1$ given by $(a, b) \mapsto (b, a)$, interchanging the S^1 factors of the torus, the resulting manifold is S^3 . To see this, consider the decomposition of $S^3 = \mathbb{R}^3 \cup \{\infty\}$ given by revolving $S^2 = \mathbb{R}^2 \cup \{\infty\}$ about some central axis $\ell \cup \{\infty\}$ (ℓ being the axis in $\mathbb{R}^2$):

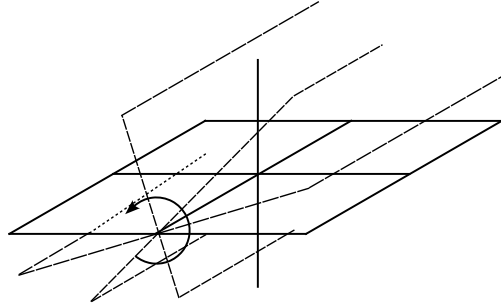


Figure 2: A decomposition of S^3 into S^2 's obtained by rotating about $\ell \cup \{\infty\}$

Consider the following subsets of one of the copies of $\mathbb{R}^2 \cup \{\infty\}$:

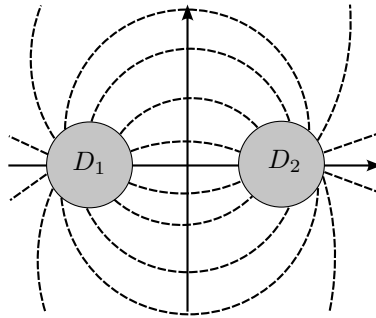


Figure 3: Subsets of S^2 inducing a genus 1 Heegaard splitting of S^3

As we rotate about $\ell \cup \{\infty\}$, the disks D_1 and D_2 sweep out a solid torus T_1 . Moreover, each of the arcs connecting D_1 and D_2 sweep out disks in S^3 . The set of all such arcs is parametrized by the central axis, which is a circle, hence $S^3 \setminus T_1$ is another solid torus T_2 , and $S^3 = T_1 \cup T_2$.

Theorem 2.1. Any closed orientable 3-manifold admits a Heegaard splitting.

Proof. The idea is as follows: take some cell structure or triangulation of a closed orientable 3-manifold M , and note that a tubular neighborhood of the 1-skeleton and its complement are both handlebodies. To see this explicitly: let T be a triangulation of M . Replace each vertex of T with a B^3 , each edge by a copy of $D^1 \times D^2$, each face of a tetrahedron with a “plate” (a copy of $D^1 \times D^2$), and each tetrahedron by a ball. Let H be the union of the balls and cylinders, and H' be the union of faces and interiors. This demonstrates very explicitly that the tubular neighborhood of the 1-skeleton of T is a handlebody, since H is a copy of B^3 with 1-handles attached. Similarly, H' is a handlebody since the plates are also 1-handles attached to 3-balls. $\square$

3 Stable Equivalence

The previous section showed uniqueness, here we will show that Heegaard splittings are also unique up to the correct notion of equivalence. In this case, this notion is *stable equivalence*:

Definition 3.1. Given a Heegaard splitting $M = H_g \cup H'_g$ of genus g , one can easily construct a Heegaard splitting of M of genus $g + 1$ called the *stabilization*. Simply add a 1-handle B to H_g such that there exists a disk bounded by the longitude of B :

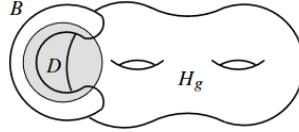


Figure 4: Stabilization

we call such a 1-handle *unknotted*. An unknotted 1-handle always exists, since we can take a small neighborhood of a point on the splitting surface to get a disk S , and consider a tubular neighborhood $\nu(S) \cong D^3$ of this disk inside M . Obviously $H_g \cup B$ is a handlebody of genus $g + 1$, since we’ve literally added a handle, and moreover $H_g \cup (D' \setminus B)$ is a handlebody if we think of $D \times D^1$ as a 1-handle attached to H'_g .

Example 3.1 The genus 1 splitting of S^3 in Figure 3 is the stabilization of the genus 0 splitting gotten by the union of two 3-balls as in Figure 1.

Definition 3.2. Two Heegaard splittings are called *equivalent* if there exists a homeomorphism of M which induces homeomorphisms on the splitting handlebodies, and are called *stably equivalent* if they are equivalent after applying stabilizations to either of them some number of times.

Theorem 3.1. Any two Heegaard splittings of a closed orientable 3-manifold M are stably equivalent.

Proof Idea. It turns out that any Heegaard splitting can be described as coming from a triangulation as we constructed in Theorem 2.1. Then given two Heegaard splittings, we can show that certain subdivisions (called *star subdivisions*) correspond to an application of a sequence of stabilizations. Moreover, any two triangulations are related by a sequence of star subdivisions, hence the result holds. The details of this proof are filled out in Saveliev. $\square$